PARTICULAR SPECIFICATION P4

Process – Mechanical Works

CONTENTS

NOTE: THIS DOCUMENT SUPERCEDES THE ORIGINAL VERSION OF THE P4 – PROCESS-MECHANICAL WORKS AT THE CLOSE OF TENDER. THE CHANGES HAS BEEN HIGHLIGHTED IN YELLOW.

P4.1 INTRODUCTION

P4.1.1 Overview

(a) This section describes the requirements for process-mechanical equipment for:

(i) Digestion pre-treatment: The pre-treatment of biosolids generated by Tuas Water Reclamation Plant (TWRP) prior to anaerobic digestion;

(ii) Sidestream Nitrogen (N) removal: The Nitrogen-removal treatment of dewatered centrates (post dilution) prior to returning to Industrial Liquids Modules (ILM); and

(iii) All systems described shall comply with the requirements of Process Mechanical General Specifications G4, except where differing requirements are given herein.

(b) The digestion pre-treatment facility will be located within the biosolids facility and will treat a portion of the overall biosolids generated by TWRP.

(c) The digestion pre-treatment facility will receive pre-dewatered sludge comprising either:

(i) Waste Activated Sludge (WAS); or

(ii) A combination of WAS and Thickened Primary Sludge (TPS).

(d) The digestion pre-treatment facility comprises a Thermal Hydrolysis Process (THP) plant, Hydrolysed Sludge (HS) cooling heat exchangers and ancillary systems (such as sludge pumping and compressed air system) as described in P4.3.1(j).

(e) The Sidestream N removal facility will be located southwest of the ILM.

(f) The digestion pre-treatment and Sidestream N removal facilities shall be designed with provision to allow expansion of treatment capacity in the future. To facilitate this, space is reserved for the expansion of the digestion pre-treatment and Sidestream N removal for future TWRP expansion as described in Sections below.

(g) The Contractor shall guarantee specific performance parameters for key mechanical systems and equipment in accordance with Volume 1, Annex O –Schedule of Process Guarantees.

(h) The Contractor shall coordinate with other contractors for the completion of the works as defined in Particular Specification P2 Preliminaries and General. In particular the Contractor shall coordinate with the biosolids facility contractor (C4A contractor).

(i) Unless otherwise stated, centrifugal pumps for sludge and/or used water shall have impellers of single-vane semi-open screw type.

P4.1.2 Hazardous Areas

(a) A Hazardous Area Classification (as per ISO IEC 60079-10) has been completed and the report is included in Particular Specification Appendix P5A – TWRP – Hazardous Area Classification. This describes the 3-dimensional areas assessed as hazardous zones from the standpoint of potentially explosive atmospheres.

(b) The Contractor shall confirm the Hazardous Area Classification (HAC) based on the selected equipment and layout. The selection of equipment and layouts shall be in accordance with the principles of safety in design:

(i) Avoid or minimise the risk of spillage or release of a dangerous substance;

(ii) Avoid the occurrence of explosive atmospheres;

(iii) Prevent ignition of dangerous substances or explosive atmospheres;

(iv) Prevent uncontrolled chemical reactions that can give rise to explosive atmospheres;

(v) Avoid the spread of fires or explosions through interconnecting plant and equipment; and

(vi) Mitigate the effect of a fire or explosion so as not to endanger people.

(c) The HAC shall be strictly adhered to in terms of suitably selected permanent, temporary and portable equipment and instrumentation. The zoned areas shall be assigned as specified in the HAC Report.

(d) All equipment (mechanical or electrical) within the hazardous areas shall be supplied, used and marked in accordance with the specified body. All such documentation associated with selection of equipment (including Declarations of Conformity and Risk Assessment details for equipment selection) shall be retained and filed along with the area classification documentation.

P4.1.3 Plant Safety

(a) The Digestion Pre-treatment and Sidestream N Removal Facilities shall comply with all Singapore Legislation, Standards and Codes of Practice.

(b) In addition to complying with the Workplace Safety and Health (Design for Safety) Regulation, an additional safety plan shall be produced and submitted demonstrating compliance and demonstrating that the principles of designing for safety have been followed.

(c) Safety Showers with eye wash shall be provided in accordance codes and standards and assessment from safety plan, HAZOP, HAC, ALMOP and DfS.

(d) A preliminary assessment of key hazards in THP has been undertaken and they are summarised in Table P4. 1. These shall be provided as minimum.

Table P4. 1: THP Preliminary Hazard Assessment

Hazard	Safety Risk	Layers of Protection	
Pressure	Injury to personnel as a result of exposure to high pressure	1	Pressure monitoring with warnings, alarms and shut down on rising pressure.
		2	Safety Relief Valve (SRV) on all pressure vessels with all discharges routed to SRV release header venting at high level. SRV on all steam pipelines with discharge routed to safety location.
		3	All pressure vessels and steam piping designed and approved as per Ministry of Manpower (MOM) regulations. Hot process pipes designed, fabricated and installed as per appropriate codes.
	Damage to equipment as a result of over-pressurisation	1	Pressure monitoring with warnings, alarms and shut down on rising pressure.
		2	SRV on all pressure vessels.
		3	All pressure vessels and steam piping designed and approved as per MOM regulations. Hot process pipes designed, fabricated and installed as per appropriate codes.
Temperature	Injury to personnel as a result of exposure to high temperature	1	All steam pipes thermally insulated; All hot surfaces (vessels and pipes) guarded against touch.
		2	All condensate and process drains routed to safe locations.
		3	All SRV discharges routed to safe locations
		4	All pressure vessels and steam piping designed and approved as per MOM regulations. Hot process pipes designed, fabricated and installed as per appropriate codes.
Foul Air	Respiratory health risk and nuisance to personnel	1	All foul air from the process, condensed and all non-condensable gases transferred to the digesters.
		2	Bursting discs upstream of SRV to prevent leakage.
The above layers of protection are in addition to signage, PPE and Safe Working Procedures for isolation, bleeding, drainage and disposal.			

(e) A Hazard and Operability (HAZOP) study shall be completed by the Contractor. The HAZOP shall be facilitated by the Contractor using independent, external qualified HAZOP facilitator subject to the approval of the S.O.. As a minimum, THP and Sidestream N Removal manufacturers' principal design engineers, process, and Instrumentation & Control (I&C) engineers, along with representatives of the Contractor shall be required to attend. All 30% design stage model and drawings (layouts, P&IDs etc.) and functional descriptions and operating philosophies shall be available for the HAZOP workshop and 30 copies of all documentation shall be delivered to the S.O. at least 2 weeks, prior to date of the HAZOP workshop. It is expected that the HAZOP be carried out over at least 3 days.

(f) An ALMOP (Access, Lifting, Maintenance and Operability) Study shall be completed during the design phase. As a minimum, THP and Sidestream N Removal manufacturers' principal design engineers, process, and mechanical engineers, along with representatives of the Contractor shall be required to attend. The Contractor shall engage an independent, external facilitator for the duration of the ALMOP. All 80% design stage model and drawings/views shall be available for the ALMOP workshop and 30 copies of all documentation shall be delivered to the S.O. at least 2 weeks, prior to date of the ALMOP workshop. It is expected that the ALMOP be carried out over at least 2 days.

P4.1.4 Purpose of the Tender Design and the Contractor's Responsibilities for Completing Design

(a) Drawings provided illustrate the S.O.'s conceptual design for the purpose of tendering. Drawings show the location, scale and interfacing requirements of the facility. Drawings shall not be taken as limiting the responsibility of the Contractor for the provision of a fully functional THP and Sidestream N removal facility.

(b) At the time of Tender, the Contractor may propose to modify the S.O.'s reference design and indicative layout to suit his choice of technology to meet the specified requirements.

(c) The Contractor shall be deemed to have included in the Contract Sum all costs for the provision of a fully functional system, whether fully itemised in the Specifications or not. The THP and Sidestream N Removal facilities shall:

(i) Meet the specified performance requirements;

(ii) Fit the space allocated;

(iii) Allow for complete system integration and safe, reliable functionality of the facility, including all ancillary plant and equipment; and

(iv) Include all provisions for safe access for operation and maintainability of all process systems, equipment, pumping systems, instrumentation, valves piping and tanks.

P4.2 OVERALL THERMAL HYDROLYSIS PROCESS (THP)

(a) This Section covers the work necessary for the provision of the THP, including auxiliary equipment for the cooling of hydrolysed sludge (HS), maintaining digester temperature, safety systems, electrical and control systems. Some of the component systems are described in more detail in latter sections.

(b) The Contractor shall provide a complete THP from the standard product range of an established THP manufacturer.

(i) The THP shall include but not be limited to, the following elements to form a fully functional and complete system:

1) THP feed (Dewatered Waste Activated Sludge (DWAS) Cake) pumps;

2) THP trains each consisting of key components, but not limited to:

a) Pre-heating stage;

b) Reactors;

c) Flash tanks / Buffer tanks;

d) THP Reactor feed pumps;

e) Process gas cooler and process gas skid; and

f) Hydrolysed Sludge Cooling heat exchangers and its recirculation pumps.

3) Hydrolysed Sludge Cooling heat exchangers and recirculation pumps;

4) Hydrolysed sludge digester feed pumps;

5) Hydrolysed sludge recirculation pumps;

6) Steam piping and Desuperheaters;

7) Plant water header, ring main;

8) Inline Steam Spargers (for Supplemental heating of Hot water supply to Greasy Waste Receiving Facility (GWRF)) and associated piping and valves;

9) Compressed Air System (For Instrument Air Supply);

10) All necessary pipework, valves and fittings;

11) A complete electrical system;

12) Control system complete with Programable Logic Controller (PLC), local touch-screen Human Machine Interface (HMI), and HMI programming, all accessories and appurtenances, control panels, instruments, cabling works, and interfaces including interface to Supervisory Control And Data Acquisition (SCADA). SCADA configuration shall be carried out by the S.O. The manufacturer shall provide the Board with full access to all aspects of the PLC code;

13) All necessary supporting steel work and reinforced concrete footings and plinths; and

14) All necessary access and maintenance platforms.

(ii) The Contractor shall guarantee the performance of the THP, including the required throughput capacity and energy efficiency in accordance with Volume 1, Annex O – Schedule of Performance Guarantees;

(iii) The Contractor shall provide all components and accessories of the system to enhance compatibility, ease of operation and maintenance, and as necessary to place the equipment in operation in conformance with the specified performance, features and functions;

(iv) The Contractor shall ensure that the THP manufacturer will be responsible for the provision of:

1) THP trains including vessels, all layouts, structural calculations for floors and walls, and vessel material selection.

2) Hydrolysed Sludge Cooling heat exchangers and associated recirculation pumps.

(v) The Contractor shall ensure that the THP manufacturer will be responsible for the provision of interconnecting piping and valves supplied as part of the THP and pre-assembled, including supporting structure and pipe supports.

P4.3 THERMAL HYDROLYSIS PROCESS

P4.3.1 Overview

(a) This section describes the requirements for the Thermal Hydrolysis Process (THP).

(b) Part of the biosolids treatment facilities at TWRP is an anaerobic digestion facility for the generated sludge, food waste and greasy waste. After digestion, the sludge is dewatered and transferred to the Integrated Waste Management Facility (IWMF).

(c) Prior to digestion, the THP shall be provided to increase the efficiency of the digestion process and enable the digesters to operate with higher feed solid content. Figure P4. 1 provides an overview to the scope of digestion pre-treatment in the Contract.

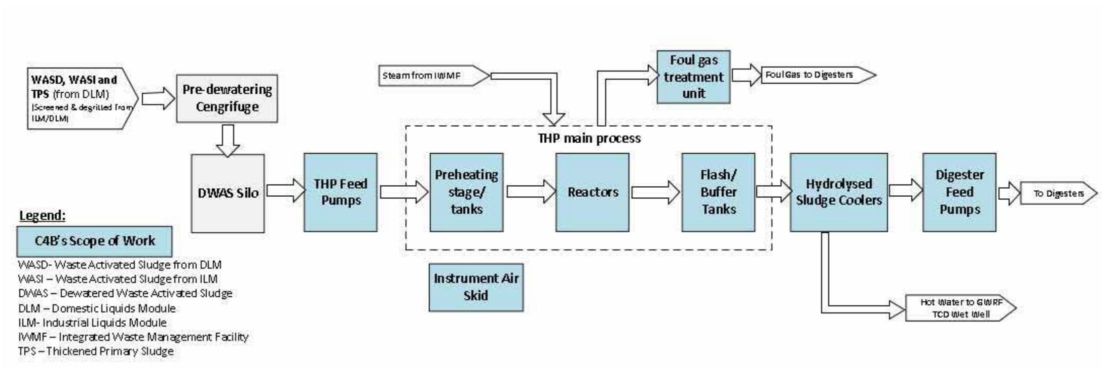

Figure P4. 1 Overall Digestion Pre-treatment process

(d) The Phase 1 Digestion Pre-treatment THP system shall process at least 50% of the solids from the liquids modules in TWRP Phase 1. All of the WAS generated from the Industrial Liquids Modules (ILMs) and Domestic Liquids Modules (DLMs) and a portion of the thickened primary sludge (TPS) from the DLMs (PSD) (due to the biosorption process employed in the DLMs) are mixed in WAS Holding Tank and then dewatered into DWAS before feeding into the THP. The addition of a portion of thickened PSD will further enhance the sludge rheology to digester feed and maximizing the use of the THP.

(e) The Digestion Pre-treatment THP system provided under this Contract is for two(2) THP trains which is two-thirds of the Phase 1 duty capacity as described in P4.3.2. The THP modules and heat exchangers of Digestion Pre-treatment Facility for Phase 1 shall occupy the C4B Working Area at the THP Courtyard as indicated in the Drawings. There will be other equipment for the Digestion Pre-treatment Facility located within the Biosolids Building, the Contractor may refer to Volume 6 for their indicative location and coordinate the final location of these equipment with Other contractors.

(f) The THP Manufacturer shall have at least eight (8) years' experiences in the provision of THP for municipal used water reclamation plants. The manufacturer requirements and information required for each previous project are detailed in Volume 1. The experience and reference projects shall incorporate the proposed process, including the principles of operation, the type and brand of key components.

(g) The THP shall be a complete system designed for thermally hydrolysing used water solids to:

(i) Reduce viscosity, allowing the downstream mesophilic anaerobic digestion to be operated at higher solids contents, at a level to achieve full digestion and prevent washout of the bacteria;

(ii) Enhance subsequent anaerobic digestion performance in terms of solids destruction and biogas yield; and

(iii) Enhance final sludge dewaterability.

(h) Thermal hydrolysis shall be achieved by heating of the sludge to a set temperature and pressure for a set time period. Each THP train shall be a semi-batch process with the overall system providing continuous feed and continuous hydrolysed sludge discharge.

(i) In addition to providing improvement of biodegradability of the sludge through Thermal Hydrolysis process, the Contractor shall design and deliver a system that also meets the other digestion pre-treatment system goals:

(i) Provide Hydrolysed Sludge cooling to achieve Digesters' bulk temperature; and

(ii) Provide Hot water supply to GWRF (and GW holding tanks) via heat recovery from hot hydrolysed sludge.

(j) The Contractor shall provide THP trains each consisting of key mechanical components, but not limited to:

(i) THP (DWAS Cake) feed pumps;

(ii) Pre-heating stage;

(iii) Reactors;

(iv) Flash tanks/ Buffer tanks;

(v) THP process pumps;

(vi) Process gas cooler and foul gas skid;

(vii) Hydrolysed sludge digester feed pumps;

(viii) Hydrolysed Sludge Cooling heat exchangers and pumps ; and

(ix) Hydrolysed sludge recirculation pumps.

(k) The Contractor shall provide THP supporting systems, but not limited to:

(i) Steam piping and desuperheaters;

(ii) Plant water header, ring main;

(iii) Inline Steam Spargers (For supplemental heating of hot water supply to GWRF) and associated piping and valves;

(iv) Compressed Air system (For Instrument Air Supply);

(v) All necessary pipework, valves and fittings; and

(vi) All necessary maintenance and access platforms, equipment supports and plinths.

(l) The Contractor shall provide key components that form the main THP are from the standard product range of an established THP manufacturer. The main THP process equipment (including the foul gas treatment unit) is shown as package units in the P&IDs and shall be the end product of one responsible THP manufacturer.

(m) To ensure compatibility and integration, the design and manufacture of the feed systems and ancillary equipment, where not supplied by the THP manufacturer, must be certified as compatible by the THP manufacturer. These systems shall include as a minimum:

(i) THP feed (DWAS Cake) pumps;

(ii) Steam supply and desuperheating system; and

(iii) Hydrolysed Sludge Digester Feed pumps.

(n) The associated upstream, downstream and ancillary services are to be integral parts of the THP and function together to achieve the required performance in a safe, efficient and stable manner. Any equipment required for the operation of the THP, including the associated upstream, downstream and ancillary services that are not shown in the Drawings and not provided by Other contractors shall be included in the design and provided at no additional cost.

(o) To assist the Contractor in understanding the required general arrangement of the THP supporting process-mechanical equipment and systems, a reference design (excluding main THP process units) has been described in reasonable detail in the Drawings and in the BIM Tender Model. However, it is the Contractor's responsibility to provide all Works to meet all the requirements in the Specifications. The Drawings and BIM Tender Model is not intended accurately to reflect the detailed design of the process-mechanical equipment/ systems and does not in any way absolve the Contractor from meeting his contractual obligations.

P4.3.2 THP Design Basis

(a) Only two (2) out of three (3) duty THP trains shall be provided for this Contract to operate in N configuration. The Contractor shall provide space and connections to retrofit the third duty train and one (1) additional THP train in order to allow the THP to operate in "N-1" configuration in the future. Ultimately, in this future configuration when one (1) THP train is out of service, the remaining number of THP units shall meet the maximum plant capacity. The number of duty THP trains installed in this Contract shall be two (2), the number including the future THP trains to form a "N-1" configuration shall be four (4). These four (4) units shall be located in the space allocated for Phase 1.

(b) The duty THP trains installed in this Contract must be able to efficiently operate at the initial conditions, at a turndown of 50% of Phase 1 flowrates.

(c) The basis of design for the THP is given in Table P4. 2. The feed parameters are further specified in APPENDIX P4D – THP Feed Parameters.

Table P4. 2 Basis of Design, THP Trains

Parameters	**Unit**	**Design basis**
Duty THP Train (for Phase 1)	#	3 NOTE: Only two (2) out of three (3) Duty trains to be provided in this Contract
Minimum Thermal Hydrolysis Temperature	°C	165
Minimum reaction time (at Minimum Thermal Hydrolysis Temperature)	min	20
Feed, DWAS (Dry Solids Content)	% DS (% Dry Solids)	16-18
Train Capacity	tDS/d (dry tonnes/day)	TBC by Contractor
Phase 1 THP Plant Nominal Maximum	tDS/d (dry tonnes/day)	Appendix P4D Inlet Feed Parameters Note [1]
Phase 1 THP Plant Design Capacity (Maximum Solids Loading)	tDS/d (dry tonnes/day)	Appendix P4D Inlet Feed Parameters
THP Train Design Capacity (Maximum Solids Loading)	tDS/d (dry tonnes/day) per THP train	Refer to Annex O [1]

Notes:
1. Governing case for Maximum Solids Loading based on TWRP Phase 1 MM (WAS + Partial DLM TPS) as described in Appendix P4D Inlet Feed Parameters

(d) Additionally, Table P4. 3 below provides a guide on the expected redundancy of the main equipment in THP. The main aim of the THP design is to achieve a system with good availability (as stated in (b) above) and minimal downtime. The Contractor's THP manufacturer may propose a different configuration that can further optimise and improve the system's availability without comprising other aspects.

Table P4. 3 Summary of Equipment Redundancy and Configuration Requirements

Parameters	THP (DWAS Cake) Feed Pumps	THP Reactor Feed Pumps (within each THP train)	First Stage HS Cooling Heat Exchanger and Recirculation Pump	Second Stage HS Cooling Heat Exchanger and Recirculation Pump	Hydrolysed Sludge (HS) Digester Feed Pumps	HS recirculation pumps[1]
Capacity Redundancy	2x100% per THP train	2x100% (per THP train	Min 2x50% per THP train Note 2	Min 2x50% per THP train Note 2	Min 2x100% per THP train	3x50% (flowrate with recirculation) 3x100% (flowrate without recirculation)
Arrangement	N+1 (1D ,1S) per THP train	To be proposed by Contractor	2 Duty sets per THP train Note 2	2 Duty sets per THP train Note 2	1 Duty and 1 Standby per THP train Note 2	With Recirculation mode: 2xDuty, 1xStandby/ Maintenance Without Recirculation: 1xDuty, 1xStandby, 1xMaintenance
Total Nos.	4 nos.	To be proposed by Contractor	4 sets	4 sets	4 nos.	3 nos.
Pump Type	Progressive Cavity	To be proposed by Contractor	Progressive Cavity Note 3	Progressive Cavity	Progressive Cavity	Screw Centrifugal or Progressive Cavity Note 3

Notes:

1. **Depending on the space utilisation of the THP Courtyard and considerations of equipment access, Contractor may propose configuration of two (2) recirculation pumps only with the same level of capacity, availability and flexibility to operate with and without recirculation flowrates. Description of HS digester feeding and recirculation are covered in P4.6.**
2. **In each train the first stage and second stage heat exchangers shall be in series as in the Drawings. The design of the overall heat exchanger configuration and capacity shall include flexibility to provide full capacity of Phase 1 hydrolysed sludge cooling when one (1) out of the twelve (12) heat exchangers (eight (8) duty and four (4) future duty) are out of service for cleaning or maintenance.**
3. **The Contractor may propose alternative pump type with relevant track record subject to S.O.'s approval.**

P4.3.3 THP Plant Layout

(a) The THP trains including the heat exchangers shall be prefabricated, modular or skid based for installation and hook-up on-site. The skid loadings shall be safely within the design loading of the THP Courtyard area as specified in loading plans provided in Volume 6 for reference.

(b) A notional arrangement based on THP process trains are shown in the BIM Tender Model for reference.

(c) Except where noted below, all THP and supporting equipment including Phase 1 future trains shall be located within the C4B Working Area at the THP Courtyard, as described in the Drawings. The layout and arrangement of the THP trains shall consider the constructability of the installation of future THP trains.

(d) The reference design includes platforms (located adjacent to edge of biosolids building) to show potential arrangement of heat exchangers installed on platforms skid structures. It shall be noted that removable railings are provided on the 12.3mSHD (L1) of Biosolids Building 's ledge for THP cooler platforms accessibility. This allows Contractor to design the platform with access from the biosolids building L1 escape walkways, which are fitted with removable railings. However, the escape walkways shall be kept clear. There shall be no obstruction from any equipment, piping or valves.

(e) The THP platforms shall be designed as standalone structures i.e., no loads shall be imposed on any part of the Biosolids Building.

(f) Also covered in other sections of the Particular Specifications, the following equipment is located outside of the C4B Working Area at the THP Courtyard as shown in Volume 6:

(i) THP feed (DWAS Cake) pumps: Located below the THP cake silos in the THP feed pump room (same level and adjacent to the south of THP Courtyard) which also house the Thickening Centrate Wet Wells and pumps;

(ii) Compressed Air System: Proposed location at Level 3 (22.6mSHD) of Biosolids Building adjacent to west of the THP Courtyard;

(iii) Low Voltage panels: In Biosolid's building LV Switchrooms 5 &6 Level 2 (15.9mSHD) of Biosolids Building north of the THP Courtyard;

(iv) Main PLC: To be located in the Process XLV and Server Room Level 3 (22.6mSHD) of Biosolids Building north of the THP Courtyard; and

(v) Associated incoming steam pipelines' instrumentations and piping items: The Contractor shall coordinate with C4A contractor on the steam pipe routing crossing through the pipe gallery room at Level 12.3mSHD of the Biosolids Building adjacent to west of the THP Courtyard.

(g) The Contractor shall take note that the fire staircase, fire accessways and escape pathways are also in the vicinity of the THP Courtyard. The Contractor shall coordinate with C4A contractor to ensure the THP layout does not cause any obstruction to the fire escape plan at the Biosolids Building as marked in the Volume 6.

P4.3.4 THP Particular Requirements

(a) The operating conditions (Temperature and Pressure) of THP reaction shall be achieved by the direct injection of live steam.

(b) Contractor shall submit the Isolation and Drain philosophy for THP Plant. As a minimum the Equipment and valve isolation shall be in accordance with the following principles:

(i) All modulating control valves shall have isolation valves for maintenance; the isolation and drain configuration shall take into account of the pressure rating upstream and downstream of the control valve;

(ii) Positive isolation (such as spectacle blinds with single or double block and bleed arrangement) shall be provided for steam and high temperature services where equipment or control valves can be taken offline for maintenance. These configurations shall be reviewed in the HAZOP study;

(iii) Removable spool pieces shall be considered for isolation where access and clearance are required for inspection or cleaning of internals components e.g. concentric pipe heat exchangers;

(iv) All manual valves over 300 mm diameter shall be actuated with positioner feedback;

(v) All valves with frequent operation (>once per week) shall be actuated and remotely operated; and

(vi) The actuation of valves can be electrically or pneumatically powered for the Digestion Pre-treatment facility.

(c) The THP shall accept DWAS (a mixture of pre-dewatered WAS and Thickened PSD) from the DWAS silos as required.

(d) When the THP is operating it shall accept a continuous or intermittent flow of DWAS from the THP (DWAS Cake) feed pumps. The THP System shall be designed to operate continuously, providing consistent and continuous HS feed to the digesters.

(e) THP shall include pumps to transfer the hydrolysed sludge to the digester HS feed ring main. The redundancy of the hydrolysed sludge digester feed pumps shall be in accordance with Table P4. 3 and such that in the event that one duty pump goes out of service, the THP train shall continue in operation with a turndown capacity without affecting the performance. Further descriptions are provided in Section P4.6.

(f) The hydrolysed sludge shall be cooled to control the digester temperature. The cooling capacity shall be designed to maintain the digesters' bulk temperature. Further requirements are provided in Section P4.5.

(g) The steam for the THP shall be supplied by the adjoining IWMF. Further descriptions are provided in Section P4.8.

(h) Any process gases arising from the THP shall be cooled before being routed to the digesters. The pressure required for the process gas to discharge into the digesters is stated in Appendix P2B, interface register. No venting to atmosphere shall be employed as part of normal operations. Condensates from process gas skids shall be routed back into THP trains.

(i) Any safety valves in contact with sludge shall be equipped with rupture discs. The safety valves shall be arranged so that they discharge safely, not into an open area, e.g. turned to discharge downwards.

(j) Pressure vessels shall be provided according to approved design codes as required by MOM regulations in Singapore. In addition the following manufacturing standards can be applied:

   (i) EN 13445 + Pressure Equipment Directive (PED) Certificate with CE marking according to PED + EU Machine directive; Or

   (ii) ASME BPVC Boiler Pressure Vessel Code Section VIII Division 1.

(k) Process Pipework shall be provided according to approved design codes in Singapore. Either of the following standards shall be applied:

   (i) EN 13480 Metallic Industrial Piping + Pressure equipment Directive (PED);

   (ii) ASME B31.3 Process Piping.

(l) Non-Building Structural Steelwork including maintenance and access platforms, access staircase, pipe rack, handrails etc. shall be provided according to:

   (i) EN 1993-1 Eurocode 3 Design of Steel Structures

(m) When built in accordance with EU applicable standards the overall plant will be supplied with complete CE marking per train and will be built in accordance with the EU Machine directive 2006/42/EC.

(n) The Contractor shall provide all necessary specialist fire and safety systems required for the THP and these shall meet the minimum requirements and any additional requirements of the approved Safety Report.

(o) Equipment access facilities, including platforms, stairways, and ladders are required to provide convenient access to all areas of equipment requiring maintenance or access to instrumentation. This is further defined in P4.16. Additionally, equipment lifting and transporting equipment and systems are required as further defined in P4.16(d).

(p) A complete instrumentation and control system shall be provided.

(q) The Contractor and Contractor's Qualified Person (QP) are responsible for obtaining all approvals, permits from the relevant authorities and agencies for the operation and maintenance of the THP.

(r) Comprehensive safety pressure management and relief systems shall be provided to protect plant and personnel and analysed in the HAZOP study. Alternative approaches to those in the Tender Documents are subject to the S.O.'s approval.

(s) The THP manufacturer shall ensure that a proven method of indirect online flow monitoring for THP DWAS feed is provided. Where dilution control of THP DWAS feed is required, the THP manufacturer shall propose the reliable method for indirect %DS measurement.

P4.4 THP FEED (DWAS CAKE) PUMPS

(a) The minimum requirement for the THP feed (DWAS Cake) pumps are provided in Table P4. 3 above and the Drawings. The detailed pump schedule shall be developed by the Contractor.

(b) As well as complying with the requirements of the General Specifications, materials of construction of THP feed (DWAS Cake) pumps shall be compatible with sludges being pumped.

(c) The pumps shall be arranged as per the Capacity Redundancy and Arrangement described in Table P4. 3 above and the Drawings, with Duty pumps drawing DWAS from DWAS silo 1 and the other Standby pumps from the DWAS silo 2.

(d) To the maximum extent feasible, long radius swept bends or tees and shall be designed with minimal fittings. The aim is to minimise frictional head loss and promote an energy efficient design as well as preventing blockages. Short or standard radius bends shall not be used.

(e) The THP feed (DWAS Cake) pumps shall be provided with inlet hopper with screw feeder. Provision shall be made for foul air (FA) extraction from inlet hoppers and connection to the biosolids odour control facility. Design of odour containment shall be in accordance with the General Specifications. A tie-in point at the nearest FA manifold will be provided by C4A contractor for connections, the location of the tie-in point will be shared after commencement.

(f) The THP feed (DWAS Cake) pumps shall be located in the THP feed pump room under the DWAS Silos as in the Drawings. The TCD Centrate wet well is located in the THP feed pump room and will incorporate four (4) pipe sleeves provided by Other contractors as shown in the Volume 6. The pipe sleeves shall be used by the Contractor to install the THP Cake Feed DWAS pipelines from the DWAS silos to the THP units.

(g) Space and connections shall be provided for THP feed pumps of the future third and fourth THP trains as shown in Volume 5 Drawings.

P4.5 HYDROLYSED SLUDGE COOLING

P4.5.1 Overview

(a) The hydrolysed sludge shall be cooled in concentric tube heat exchangers prior to being fed to the digesters.

(b) Heat exchangers shall be designed and sized for the full range of operating duties, temperatures and conditions specified.

(c) The heat exchangers shall be of proprietary design and shall be from the standard product range of a specialist heat exchanger manufacturer with a track record of reference facilities in successful operation on similar sludges and with similar duties and capacities.

(d) The scope of Hydrolysed Cooling system shall include the provision of heat recovery for hot water supply to GWRF during operating hours, and diversion to wet well during non-operating hours. This shall include a set of steam spargers, related control valves and instrumentation to provide supplementary heating when heat recovery through Hydrolysed Cooling is not available. Further details are covered in Sections below.

P4.5.2 Hydrolysed Sludge Cooling and Heat Exchanger Arrangement

(a) It is envisaged that the Hydrolysed Sludge has to be cooled to approximately 50-65°C, in order to maintain the Digesters' temperature at 36°C or a setpoint between 36-40°C. The hydrolysed sludge from the THP units shall be cooled by a series of heat exchangers. In order to keep the hydrolysed sludge at the required solid content, cooling by means of dilution shall be avoided. The requirement of Phase 1 hydrolysed sludge cooling is summarised in Table P4. 4, Table P4. 5 and Table P4. 6.

Table P4. 4 Hydrolysed Sludge Cooling Requirements

Descriptions	**Units**	**Value**
Hydrolysed Sludge (Phase 1 MM, Maximum Month Flow) [1]	m^3/d	1,461 (@ 13.6%DS)
Hydrolysed Sludge (Phase 1 AA, Annual Average Flow) [1]	m^3/d	1,107 (@ 13.6 %DS)
Turndown ratio	%	50%
Type	-	Concentric tube
Configuration	#	2 stages (in series) dedicated to each THP train;
First Stage HS Cooling Heat Exchangers:	-	Maximize heat recovery to provide hot water for GWRF while providing partial cooling of Hydrolysed Sludge
Second Stage HS Cooling Heat Exchangers:	-	Ensure the additional cooling of hydrolysed sludge required to maintain a hydrolysed sludge temperature setpoint
Cooling Water inlet temperature	°C	29 to 32
Notes: 1. Actual volumetric flowrate of Hydrolysed Sludge from THP shall be based on the THP manufacturer's heat and material balance. Peak hydraulic flowrate to be confirmed by THP manufacturer considering any recirculation pumps in between the heat exchangers as required.		

Table P4. 5 First Stage HS Cooling Heat Exchangers

Descriptions	Units	Values
Hot Medium (Hydrolysed Sludge) temperature	°C	In: 105-115 (To be designed by THP manufacturer) Out: (Intermediate temperature To be designed by THP manufacturer)
Minimum Cooling water outlet (HWR, Hot Water (Return)) temperature (Required for GWRF's LPHW)	°C	50
Hot Water outlet flowrate required (for GWRF)	m^3/h	105 [1]
Estimated Heat Recovery required	kW	2797 [2]

Notes:

1. The HWR flow controller at the combined outlet of the First Stage HS Cooling Heat Exchangers shall maintain the HWR flow setpoint. It is envisaged that the controller and dump control valve will be designed to allow a 5-10% buffer for heat exchanger fluctuations. The Process Control Narratives (PCN) in Appendix P6A explains the required control for Hot Water supply to the GWRF. Appendix P4E also provides details on the LPHW requirements.
2. Heat Exchanger's Cooling Duty shall be confirmed by the Contractor's THP manufacturer

Table P4. 6 Second Stage HS Cooling Heat Exchangers

Descriptions	Units	Values
Hot Medium (Hydrolysed Sludge) temperature	°C	IN: Intermediate temperature TBC by THP manufacturer OUT: 50 – 65, Min 50 (Setpoint adjusted based on digester temperature controller)
Maximum Cooling water outlet temperature	°C	45 [1]
Estimated Cooling Duty required	kW	1970 [2]

Note

1. The hot water return is going to the TCD wet well. Maximum temperature shall be maintained below 45°C.
2. Heat Exchanger's Cooling Duty shall be confirmed by the Contractor's THP manufacturer.

(b) The cooling system shall have a minimum of two-stage of heat exchanger, arranged in series. The Contractor is allowed to propose alternatives arrangement with similar degree or improved availability and functionality of the system.

(c) As a minimum, each THP train shall have a duty / duty cooling system as described in P4.3.2 Space and connections for the 5th and 6th, 7th and 8th duty cooler trains for future THP trains 3 & 4 respectively, shall be provided. The overall common system including the hot water supply header to GWRF as described in shall be designed to be adequate for the Phase 1 capacity.

(d) Each heat exchanger shall be provided with a recirculation pump to maintain high speed, turbulent flow through the heat exchanger. Materials of construction of recirculation pumps shall be compatible with sludges being pumped.

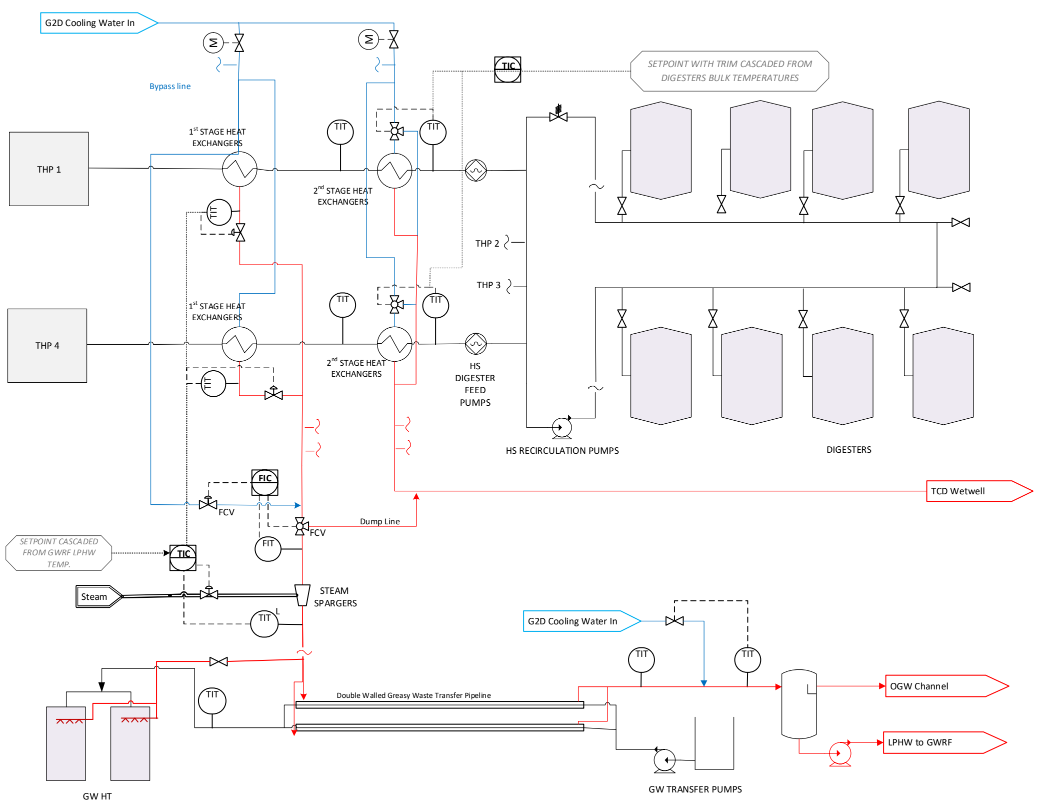

Figure P4. 2 Schematic of Heat Exchanger and Hot Water Supply to GWRF

P4.5.3 First Stage HS Cooling Heat Exchangers

(a) The objective of the first stage HS cooling heat exchanger is to maximise the heat exchange between Hydrolysed Sludge and the cooling water in order to provide a consistent hot water supply to GWRF (and Greasy Waste holding tanks). It will be used to maintain the skin temperature of the greasy waste (GW) transfer pipes and provide spray/washdown water for the GWRF.

(b) The Hydrolysed Sludge will be cooled to an intermediate temperature before entering the second stage HS cooling heat exchanger.

(c) After passing through the heat exchangers, the cooling water output from the first stage shall be controlled to a set temperature and shall be combined in the hot water supply header. The hot water supply header to GWRF shall be provided with supplementary steam heating and bypass controls as described in Section P4.5.6., to ensure that hot water supply to GWRF is not disrupted in the event when there is inadequate heat recovery from first stage HS cooling heat exchanger.

(d) The basis of design for the first stage HS heat exchangers shall be in accordance with Table P4. 5 First Stage HS Cooling Heat Exchangers above.

P4.5.4 Second Stage HS Cooling Heat Exchangers

(a) The objective of the second stage HS cooling heat exchanger is to cool the hydrolysed sludge to the required outlet temperature setpoint which shall be controlled by cascading feedback from the digesters temperature. The temperature control valve shall be provided on the cooling medium stream to maintain the required outlet temperature of the hydrolysed sludge. It shall play an important role to maintain the digesters temperature at the optimum setpoint. Further description of digester temperature control is covered in Appendix P6A Digestion Pre-treatment and Sidestream Nitrogen Removal PCN.

(b) The basis of design for the second stage HS heat exchangers shall be in accordance with Table P4. 6.

P4.5.5 Heat Exchanger Rating and Design

(a) The heat exchanger design may incorporate features on the tube side such as dimpled or corrugated tube to promote turbulence flow region thus enhancing heat transfer efficiency and resulting in optimised pumping energy requirements. If tube features are proposed for heat transfer enhancement, they shall not compromise on the quality and design life of the tube material.

(b) An allowance for the fouling of the heat exchangers shall be used in the sizing of the heat exchangers. Plant Water (G2D) is the cooling water. Refer to Appendix P4A for the Plant Water (G2D) characteristic for estimation of fouling factor on the cooling water side.

(c) The materials of construction of heat exchangers shall be compatible with sludges being pumped.

(d) The U-bends of the inner pipe shall be easily removable to provide flushing and rodding/jetting points to enable the quick clearance of any blockages and to clean the pipes. Pipework shall be provided to enable the flushing water to be directed to the drainage system. The Contractor shall provide a method for heat exchanger cleaning, either high pressure jet cleaning or chemical cleaning shall be proposed. Notwithstanding this, the heat exchanger shall be designed with mechanical integrity to allow high pressure jet cleaning, with maximum jetting pressure specified by the THP manufacturer.

(e) All hot surfaces within reach of personnel shall be guarded to prevent contact. Suitable guard such as removable wire mesh guard or thermal insulation shall be provided at all pipework within reach of personnel to prevent accidentally touching hot surfaces. In addition, safety signages for hot surfaces shall be provided for at appropriate locations. The signages shall be submitted for S.O.'s approval.

(f) The Contractor shall be responsible for the thermal design and rating of the heat exchanger. The design conditions of both cold and hot streams shall be considered in the thermal design. Thermal design calculation for Heat Exchanger rating shall be submitted for approval.

P4.5.6 Hot Water Supply Control to GWRF

(a) The tie-in for hot water supply to Greasy Waste Receiving facilities are described in Appendix P2B.

(b) The following describes the requirement of Hot Water to GWRF (As described in Appendix P4E).

(i) Flowrate: 105m^3/h at minimum temperature of 50°C (During GWRF operating hours: 8am to 6pm from Mon to Sat). The tuning of the temperature setpoint shall be cascaded from GWRF arrival temperature which is expected to be approximately 40-45°C.

(ii) During non-operating hours, the hot water from first stage shall be diverted to TCD wet well. The control of the modulating valve should still accommodate for the intermittent hot water spray demand from Greasy Waste Holding Tank.

(iii) Interface pressure shall be minimum 4barg.

(c) All hot water supply pipework between the heat exchangers and the interface point shall have thermal lagging.

(d) The Contractor shall provide temperature control for the cooling water discharge from each first stage heat exchanger. The minimum valve and instruments required to enable temperature control is shown in the Drawings.

(e) It is envisaged that the demand for hot water would be interrupted during THP turndown operation or when any trains are taken out of service.

(f) A set of steam spargers shall also be provided after the first stage HS cooling heat exchangers to provide supplementary heating when heat recovery from Hydrolysed Sludge Cooling process is not available or not sufficient to meet the GWRF demand. Further description on steam spargers is provided in P4.8.5.

(g) Common bypass control valves shall be provided around the first stage HS cooling heat exchangers. It is envisaged that in the event of one or more first stage HS cooling heat exchangers are unavailable; the common bypass flow control valves will modulate and provide required make-up flowrate based on the combined hot water supply flowrate feedback. The resulting low temperature alarm shall initiate the start of inline heating via steam spargers.

(h) The Contractor shall carry out pipeline simulations with scenario checks on the interconnecting hot water supply lines to ensure no water hammer or other, high or low, pressures due to surge occur (at the hot water supply line interface and downstream piping), as a result of the selected heat exchangers configurations and control valves.

(i) The Contractor shall be responsible for proper control valve selection and controls to ensure stable interface conditions and to ensure the required hot water operating conditions via the hot water supply line to GWRF.

(j) The Contractor shall be responsible for proper selection and determination of relief valves, thermal expansion tank and equipment, or other overpressure protection measures to prevent any potential damage to the plant and equipment or risk to safety.

(k) Overpressure scenario due to steam sparger control valves failure shall be assessed to protect downstream piping.

P4.6 HYDROLYSED SLUDGE DIGESTER FEED ARRANGEMENTS

P4.6.1 Overview

(a) The Contractor shall provide a HS digester feed pump system with:

(i) A consistent flow of HS to each of the digesters; and

(ii) A stable flow to provide stable control and accurate distribution and measurement of the feed.

P4.6.2 Digester Feed Lines

(a) The HS digester feed pumps discharge into a feed ring main, provided by Other contractors, with sub-headers and actuated valves, which operate sequentially to distribute the flows to each of the digesters, according to a pre-set cycle further described in Appendix P6A Digestion Pre-treatment and Sidestream Nitrogen Removal PCN.

(b) The HS is transferred via two (2) number of 6” diameter pipes which forms a ring loop, provided by Other contractors. It is envisaged that the hydrolysed sludge digester feed should operate in a recirculation loop mode to avoid settling out of solids along the feed line.

(c) The Hydrolysed Sludge digester feed flowrate from THP trains to the digesters shall be stable and continuous as per the required flowrate. The Contractor and the THP manufacturer shall carefully design the Hydrolysed Sludge digester feed and recirculation pumps rating and arrangement required to provide digester feed pressure for all scenarios and operate in a recirculation and other modes in order to minimise downtime.

(d) Further interface data on the HS digester feed pipework will be provided after contract award. For continuity of design, the Contractor shall coordinate with Other contractors to evaluate requirements on the interface pipework. To ensure compatibility and integration, the digester feed line and feed controls must be reviewed as compatible by the THP manufacturer.

(e) The Digesters will also be fed with Thickened Primary Sludge (TPS), Food Waste (FW) and Greasy Waste (GW). These feed systems are provided by Other contractors. The process control narrative of the digester feeding is provided in Appendix P6A Digestion Pre-treatment and Sidestream Nitrogen Removal PCN.

(f) Space and connections shall be provided for the HS digester feed pumps of future THP trains 3 and 4 as shown in Volume 5 Drawings. The common HS digester feed header, HS recirculation pumps and HS pipe interfaces to digesters shall be provided based on the Phase 1 design capacity.

P4.6.3 Pumping System

(a) The detailed pump schedule shall be developed by the Contractor.

(b) Materials of construction of HS digester feed pumps and HS recirculation pumps shall be compatible with operating temperature and fluid conveyed. The selected pump head shall take into account of the rheology of sludges being pumped (during start-up and normal operation). Pump material shall be designed to withstand the process design temperature, and safeguarding system such as thermal protection device shall be provided as per General Specifications.

(c) As a minimum the HS digester feed pumps shall be configured as described in Table P4. 3. The HS digester feed pumps shall transfer the HS to the HS digester feed ring main. Based on the available pipe routings in the Drawings, the THP contractor shall carry out hydraulic calculation to ensure the selected pump head is able to deliver to the furthest digester without the recirculation pump online.

(d) The configuration of the HS recirculation pumps shall be provided to deliver a simple yet energy efficient system with high availability. Provision of dilution or flushing line of hydrolysed sludge digester feed piping shall be included if deemed important for THP start-up and to achieve a trouble-free operation.

(e) A bypass around the HS recirculation pumps shall be provided.

(f) The discharge pipe work shall be fitted with guards or suitable coatings to prevent any contact with the hot pipes and fittings.

(g) Flushing and rodding/ jetting points shall be provided to enable the quick clearance of any blockages and to clear the pipes. Outlets shall be provided to enable the flushing water to be directed to the drainage system.

(h) Gas release valves shall be provided at the highest point(s) and shall be suitable for wastewater sludge applications. The vent gas shall be piped to an area for safe discharge. Each of the gas release valves shall include an isolation valve and access shall be provided to maintain the valves.

P4.7 PLANT DRAINAGE AT THP COURTYARD

P4.7.1 Overview

(a) A plant drainage inspection chamber will be provided by Other contractors in the corner of the THP Courtyard area (Phase 1). The Contractor shall direct spillages and drains from the digestion pre-treatment facility to these inspection chambers. Due to the fire escape corridor crossing between THP modules and the plant drainage inspection chamber, there is a main plant drain header adjacent to the THP Courtyard which will also be available for tie-in from courtyard through the apron slab for process drain from THP pressure vessel and modules, the Contractor shall coordinate and establish the tie-in location with Other contractors and S.O. prior to commencement of work.

(b) The Contractor can, subject to the approval of the S.O., request that additional connection points to the plant drainage network provided by Other contractors. The cost related to identification and connection of plant drain tie-in works shall be included in the design and deemed included in the tendered sum.

(c) The plant drainage inspection chambers discharge by gravity to the biosolids facility plant drainage pumping station.

(d) Any discharges into the plant drain network shall be at a maximum temperature of 45°C.

(e) The plant drainage system provided under this Contract shall be sized adequately for Phase 1 design capacity. Plant drains interface connections from future THP trains 3 and 4 to common plant drain header shall be provided.

P4.8 STEAM AND CONDENSATE SYSTEM

P4.8.1 General

(a) This section shall be read in addition to G4.2 Pipework, Appendix P4B and P4C for material selection. The pressure rating of the steam and condensates piping shall be carefully selected, taking into the consideration of the system design pressure and specification breaks. The piping design, testing and commissioning procedures shall comply with OSHD – Steam Piping Guide (MOM regulation).

(b) The steam is supplied from IWMF at the tie-in location on the bridge between IWMF and Biosolids Building as described in other sections. The estimated operating conditions of the supplied steam are as per table below:

Table P4. 7 Steam Supply Parameters

Steam Supply Parameters	**Conditions**	**Units**
Supply Temperature	Up to 19°C superheat (±5°C) at supply pressure	°C
Supply Pressure	1150 (Min/Max TBC)	kPag
Maximum flowrate[1]	10	t/h
Normal flowrate[1]	7	t/h
Minimum flowrate[1]	3.5	t/h
Notes 1) The steam flow rates are preliminary based on Phase 1 steam requirement and shall be confirmed by the Contractor during detailed design of THP. The Contractor shall ensure that the final design of steam supply piping shall be take into account of any flow fluctuations associated steam flow control to each THP train.		

(a) The steam pipeline from IWMF to THP shall be designed with insulation /jacket for heat conservation. This serves to maintain the quality of steam arriving to THP units, minimise the generation of condensates, as well as for personal protection. The insulation design shall be optimised to minimise risk of corrosion under insulation along the piping. The piping specifications, insulation thickness and material selected with calculations shall be submitted for S.O.'s approval.

(b) The steam supply header and condensate collection system provided under this Contract shall be sized adequately for THP Phase 1 design capacity. Interfaces to future THP trains 3 and 4 shall be provided. Pipe size for steam service shall be optimised for design flow and turndown flow. Oversizing will create higher heat loss and greater operating expenses, especially during turndown conditions.

(c) Table P4. 8 below provides a guideline for steam service.

Table P4. 8 Gas Velocities Guideline for Steam Services

Steam Services	**Velocity Limit (m/s)**	**Typical pressure drops range (bar/100m)**
Steam >3.5barg Headers/manifolds	-	0.1-0.35
Steam >3.5barg Offtakes	-	0.35-0.9
Saturated Steam	15-30	-
Flash Steam (Condensate return line)	20	-
Superheated Steam	30	-

(d) Branch lines or take-offs from main header shall be from top of the pipeline. Care shall be taken to design where high pressure drops or velocities across orifice or nozzles. Adequate provision on piping support & insulations shall be provided at critical locations to ensure that noise/vibration are safely within limits.

(e) A properly designed steam piping shall be water hammer free during start-up and normal operation. The Contractor shall include any pipeline simulation required based on the steam flow control scenarios, to ensure adequate overpressure protection due to transient scenarios or fluctuations in normal operations etc. Suitable pipeline protection measures shall be provided based on the results of the analysis to prevent any potential damage to the plant and equipment or risk to safety.

(f) In addition to personnel protection on hot pipes, safety signages shall be provided for live steam pipelines as well hot water/condensate pipes at appropriate locations. The signages shall be submitted for S.O.'s approval.

P4.8.2 Condensate Trap

(a) Steam or condensate traps shall be provided on steam lines at regular intervals (about each 30 to 50 m) and at each point where condensate could collect (such as low points). As a minimum requirement, strainer and condensate traps shall be provided upstream of control valves or pressure reducing valves in steam services.

(b) The condensate trap shall be an automatic self-adjusting device that control the discharge of condensate and non-condensables without the loss of live steam. Drip legs shall be considered for connection from steam piping to condensate trap. The type of condensate trap selected shall be fit for purpose, leak-free and energy efficient.

(c) Steam traps shall be installed in a drain pocket on the steam pipework. The trap pocket shall be sized to allow condensate to settle when steam is flowing at high velocity, and to allow space for any dirt and scale to settle. The drain pockets shall be provided in the following sizes as a minimum.

(i) For up to 100mm main diameter:

1) Pocket diameter the same as the main;

2) Minimum pocked depth 100mm.

(ii) For greater than 100mm main Diameter:

1) Pocket diameter of 100mm;

2) Minimum pocket depth of 150mm.

(d) All condensates from steam piping and condensate traps shall be returned to suitable process within THP train (such as a condensate recovery drum) and shall not be discharged to drain.

(e) Steam pipework shall be installed at a minimum 1:100 gradient. The fall shall be in the same direction as the steam flow. Where this is not possible, a combination of a steeper gradient, more frequent steam traps and lower velocity steam shall be provided to ensure adequate removal of condensate.

(f) All branch connections shall be taken from the top of the main steam pipework.

(g) A strainer shall be installed before any key equipment, including any flowmeters and regulating valves. The strainer shall be orientated with the basket on the side to prevent accumulation of condensate.

P4.8.3 Safety Valves

(a) Safety valves on pipelines and Pressure Vessels shall be provided in accordance with the respective Pressure Vessel code as described in P4.3.4 or ANSI/ASME PTC 25 for piping. Sizing of Safety Relief Valves and bursting discs shall be based on the governing relief scenario which shall be determined by the Contractor in accordance with the main national and international standards such as API 520 RP/ASME/PED.

(b) Any safety pressure relief pipework shall be routed to a safe location away from any potential operating areas.

P4.8.4 Steam Desuperheater

(a) Steam desuperheater system consisting of inline nozzle, control valves, instrumentation and controls shall be provided if required to meet the steam conditioning requirement of the THP for effective heat transfer. It is expected that each steam incoming line will have a dedicated steam desuperheater. The steam inlet piping design shall include any pressure control valves required upstream of Steam Desuperheater and THP system.

(b) RO permeate (ROP/G1) is available as cooling water for the steam desuperheater and the interface details and quality are provided in Appendix P2B and Appendix P4F. The location of the tie-in connections is indicative and the Contractor shall coordinate with Other contractors and the S.O. for final tie-in locations. Typically, the cooling water used for desuperheating will be injected at pressure higher than the steam pressure into the process line. The Contractor shall propose and provide necessary boosting pump and equipment to supply a fully functional steam desuperheater. A minimum of 100% redundancy shall be provided for booster pumps.

(c) The steam desuperheater system shall be designed by the Contractor based on the THP desuperheating requirement and with objective to minimise steam consumption and optimise water usage under various loads and turndown operation scenarios. The steam desuperheater shall meet the turndown requirements of the THP.

P4.8.5 Steam Sparger (In-line Instantaneous Jet heater)

(a) As mentioned in P4.5.6, supplemental heating with steam is required as backup when heat exchange from Hydrolysed Sludge is not available for producing hot water to GWRF. This is envisaged during initial operations and also possibly when there one or more heat exchangers are taken out of service.

(b) As such, a fully functional inline steam sparger or instantaneous jet heating system with steam shall be provided at the main supply line of hot water to GWRF. The steam supply shall be taken from the main header downstream of the desuperheater and flowmeter. Appropriate pressure let down valve configuration shall be provided based on the steam sparger supplier requirement.

(c) The steam sparger shall be designed and supplied as a complete package consisting of inline steam injector, control valves and integrated control system which promotes efficient mixing of steam and the process fluid to be heated. The arrangement indicated in the Drawings are indicative only.

(d) The steam sparger should function seamlessly during design flow and turndown load, i.e., free from steam cavitation.

P4.9 COMPRESSED AIR SYSTEM FOR THP AND BIOSOLIDS BUILDING

P4.9.1 Overview

(a) The Contractor shall provide a compressed air system, located within the Biosolids Building, to supply any plant requiring compressed air within this Contract. The compressed air system shall also be capable of providing compressed air to the biosolids facility (by Other contractors).

(b) A space for the compressed air system has been provisioned at L3 of Biosolids Building. The location and skid dimensions are indicative only, the Contractor shall coordinate with C4A contractor and S.O. on the final and approved skid and tie-in point locations.

(c) Each compressed air system shall be provided with air compressors, after cooler, water separators, particle filters, air dryers, air receivers and distribution pipework and all associated instrumentation and controls to meet the needs of the entire plant. The Contractor shall provide a fully redundant compressed air system including all accessories, actuated valves and instrumentation and controls.

(d) The compressed air system shall have a central master controller to optimise the starting and stopping of the all the machines for maximum energy efficiency.

(e) The Compressed air system shall be designed to supply instrument grade compressed air to Phase 1 THP trains (including future THP trains 3 and 4), any other systems supplied by the Contractor and the equipment in the biosolids facility being provided by Other contractors. The compressed air system shall be provided with at least +20% margin on the design capacity. The consumers in the Biosolids Building are (but not limited to):

(i) Polymer bulk bag discharger slide gates and dust extractors;

(ii) Polymer and Magnesium Hydroxide Silos outlet rotary valves;

(iii) Polymer bulk bag discharger massager system; and

(iv) Sludge cake, screenings and grit hopper discharge slide gates.

(f) Compressed air system shall provide air supply to C4A not later than the key project milestone described in Particular Specification P2 Preliminaries and General.

(g) The compressed air requirement for the equipment in the biosolids facility is shown in Table P4. 9 below.

Table P4. 9 Biosolids Building Compressed Air Requirement

Description	Unit	Value
Compressed air Grade	-	Instrument Air (IA)
Minimum Pressure	barg	8.5
Maximum Instantaneous flow rate [1]	Nm^3/hr	Refer to Appendix P2B
Maximum Daily Flowrate [1]	Nm^3/hr	15
Minimum Air Receiver Hold-up time [2]	minutes	15
Notes: 1.Preliminary estimate of Instrument air demand from Biosolids Building's consumers by Other contractors is indicative only and subject to change. The Contractor shall request for the final instrument air demand from C4A contractor for detailed design calculations of the compressed air system upon commencement. The above consumption does not include the instrument air demand from the process in this Contract. 2. Time for the receiver to go from upper to lower pressure limits at maximum daily flowrate		

(h) The Contractor shall submit the sizing calculations for air compressors and all associated components for S.O.'s approval prior to procurement.

(i) The Contractor shall carry out HAZOP, ALMOP and all necessary studies to assess safety, accessibility, maintainability and operability of the supplied plant to S.O.'s satisfaction.

P4.9.2 System Components

(a) Each compressed air system shall comprise of a minimum of two (2) oil-free rotary screw compressors, two (2) after coolers, two (2) water separators, two (2) particle, two (2) air dryers (one duty, one standby), and two (2) air receivers, in duty and standby configuration and one (1) separate master controller.

(b) Compressors shall be designed and supplied as a complete package with all necessary equipment, including but not limited to the following components: inlet filter, air compressor elements, drive motor, starter, and regulation and control system.

(c) Compressors shall be provided with the acoustic enclosure to meet the noise level specified in General Specifications.

(d) Acoustic enclosures shall be provided with ducted exhaust system for heat dissipation. All exhaust from compressor shall be ducted to outside the building or a safe location and shall not be facing any other buildings or walkways or fresh air intakes.

(e) The Contractor shall provide duty and standby water separators complete with isolation valves, auto drains with fixed piping for safe discharge to plant drain system and instrumentation required to generate alarms indicating the need for maintenance.

(f) The Contractor shall provide duty and standby particle filters complete with isolation valves and the instrumentation required to generate alarms indicating the need for maintenance.

(g) The Contractor shall provide duty and standby desiccant type air dryers.

(h) The Contractor shall provide air receivers complete with pressure switches required for operation of the compressors and monitoring from Monitoring and Control System, pressure relief valves, manual and auto-drains with fixed pipe work to plant drain system.

P4.9.3 Distribution System

(a) The Contractor shall provide the pipework, valves, fittings and all related ancillaries for a fully functional system.

(b) The Contractor shall provide a ring type distribution main for the equipment located in the THP courtyard with adequate isolation valves to facilitate continued operation of the plant in the event of failure of a section of the pipework.

(c) Position of valves shall be such that no more than one serving duty unit in any process unit shall be out of service during repairing of the distribution pipework.

(d) The Contractor shall provide secondary receivers or locally mounted compressed air systems where it is not practical to meet the air demands of plant/systems located remotely or recommended by the various equipment vendors.

(e) The Contractor shall provide required instrumentation for remote monitoring of the compressed air system flow and pressure. For instrument air grade, online dew point analysers shall be provided for monitoring the instrument air quality.

P4.9.4 Master Controller

(a) The master controller shall be enclosed in a separate wall-mounted panel with a user-friendly operator interface.

(b) The master controller shall have communication interface to the Plant SCADA system for remote monitoring and alarms. The interface shall be Ethernet based MODBUS TCP communication protocol otherwise protocol converter is required.

(c) Signals available via the interface shall include:

(i) Status of each compressor;

(ii) Current local pressure setpoint;

(iii) Current local pressure measurement; and

(iv) Current alarms and warnings for each compressor.

(d) In the event of the failure of the master controller, it shall be possible for the compressed air system to continue running automatically, albeit without optimum energy efficiency.

P4.10 SIDESTREAM NITROGEN (N) REMOVAL FACILITY

P4.10.1 Overview

(a) This section covers the requirements of the Sidestream Nitrogen (N) Removal Facility.

(b) The TWRP biosolids treatment facility incorporates a reference design for sidestream treatment of return liquors to achieve effluent total nitrogen targets. The design of the sidestream N removal facility shall be proposed by the Contractor. Partial nitritation / anammox (deammonification) is the preferred approach for the treatment of sidestream flows.

(c) There are a number of technologies suitable to achieve the performance objectives for N removal. The layout and design of the sidestream N removal facility to incorporate such technologies differs. As a result, the Contractor shall be responsible for the design-build of the Works. Aside from the selection of technology and layout, all elements of the sidestream N removal facility shall comply with the relevant Particular and General Specifications.

(d) The Contractor shall appoint a single technology provider to provide a complete sidestream N removal facility as set out in the Specifications to enable the quality and capacity requirements to be achieved.

(e) The sidestream N removal technology provider shall have experience in the provision of anammox based sidestream treatment processes for municipal Water Reclamation Plants (WRP), or similar applications. The experience requirements and information required for each reference project is detailed in Volume 1. The experience and reference projects shall incorporate the proposed process, including the principles of operation, the type and brand of key components.

(f) Civil outline drawings of the sidestream N removal facility shall be provided by the Contractor for the approval of the S.O.. The Contractor shall provide structural design for the plant and obtain BCA approval for the structural design.

(g) All detail designs, government authorities' submissions, supply, installation, construction, arrangement for authorities' inspections, testing and commissioning, including full performance testing shall be provided by the Contractor.

(h) The Contractor shall be responsible for detailed design, minimum level of standby units/systems, minimum level of instrumentation as prescribed in the P&ID and described in the Drawings and Specifications.

(i) The Contractor shall substantiate design proposals by calculations. Once the S.O. has reviewed the Contractor's calculations in this respect and has agreed the final layout, the Contractor shall carry out the detailed design of the all the process units and ancillary works.

(j) For mechanical, electrical and instrumentation works, the Contractor shall select equipment of the type, quality, accuracy and number specified, and shall substantiate the selection by submitting detailed design calculations and drawings to the S.O. for the proposed duty of the equipment proposed.

(k) Instrumentation and Control work shall meet the requirement of General Specifications G6 – ICA and Particular Specification P6 ICA Works.

P4.10.2 Sidestream N Removal Facility Design and Performance Objectives

(a) The sidestream N removal facility shall treat the following flow streams:

(i) Digested Sludge Dewatering Centrate (DC);

(ii) IWMF Dryer Condensate (SC); and

(iii) The thickening centrate returns is available for blending, and comprise of the following:

1) PSI Thickening Centrate (TCI) and WAS Pre-dewatering Centrate (DCW); and

2) Depending on operation scenario the stream may also include PSD Thickening Centrate (TCD).

(b) The Digested Sludge Dewatering Centrate (DC) and IWMF Dryer Condensate (SC) will be pumped by Other contractors from DC wet well in Biosolids Building to Sidestream N removal facility. The pumped fluid has a maximum operating temperature of 40°C. The Contractor shall provide the interconnecting DC pipeline as shown in the Drawings.

(c) Thickening Centrates (TC) are also pumped by Other contractors from respective thickening centrate wet wells in Biosolids Building via a single pipeline to ILM. The Contractor shall provide the interconnecting TC pipeline as shown in the Drawings and take-off from the main TC return line for blending.

(d) The parameters and characteristics of feed flows are detailed in Table P4. 10.

Table P4. 10 Basis of Design: Sidestream N Influent Characteristics

Parameter	Unit	Dewatering Centrate (DC)[2, 3] MM[1]	Dewatering Centrate (DC)[2, 3] AA[1]	IWMF Condensate MM[1]	IWMF Condensate AA[1]	TCI and DCW available for dilution[3] MM[1]	TCI and DCW available for dilution[3] AA[1]
Flow	m^3/d	6,052	4,895	536	428	27,942	21,235
TKN	kgN/d	10,366	7,566	0	0	1,668	954
Ammonia	kgN/d	9,651	7,084	715	435	225	107
Nitrate	kgN/d	0	0	0	0	258	167
TSS	kg/d	15,652	12,567	0	0	25,334	19,246
COD	kg/d	11,325	7,367	0	0	34,253	18,557
BOD	kg/d	1,048	722	0	0	7,481	4,078
Alkalinity	kg/d	26,041	19,400	0	0	1,746	1,916

1. MM = Maximum month; AA = Average Annual
2. The TP concentration of Dewatering Centrate is anticipated to be in the range of 100 to 500 mgTP/L.
3. The maximum Ortho-Phosphate concentrations are expected to be:
    a. Dewatering Centrate: 170mgPO_4-P/L;
    b. Blending Stream:7mgPO_4-P/L

(e) The design shall include means for DC to be diluted with the thickening centrates return stream which typically is a mixture of TCI and DCW. The design shall be based on a dilution ratio of "TCI and DCW" to DC of at least 1:1. A lesser ratio can be proposed as an alternative offer if proven in the Reference Plants downstream of THP facilities as set out in Volume 1 Annex B3.

(f) In optional operating scenarios, the blending stream may include a mixture of DCW, TCI, TCD, hot water returns and potentially small amount of acidic leachate from biosolids odour control facility. The temperature of the blending stream will be 28 to 32°C.

(g) The sidestream N removal facility shall be designed to accept and treat the feed stream envelope in Table P4. 11.

Table P4. 11 Basis of Design: Sidestream N Influent Characteristics

Parameter	Unit	Side Stream-N Feed MM[1]	Side Stream-N Feed AA[1]	Comment
Flow	m^3/d	12,640	10,219	= (DC) + (1:1 Dilution of DC with DCW + TCI) + (IWMF Condensate)
TKN	kgN/d	10,727	7,786	
Ammonia	kgN/d	10,415	7,544	
Nitrate	kgN/d	56	39	
TSS	kg/d	21,139	17,003	Instantaneous
COD	kg/d	18,744	11,645	
BOD	kg/d	2,668	1,662	
Alkalinity	kg/d	26,419	19,841	

MM = Maximum month; AA = Average Annual

(h) As a minimum, the design for the sidestream N removal facility shall include the following aspects, which shall also be deemed to be performance criteria:

(i) Performance – comply with the requirements set out in Volume 1, Annex O - Schedule of Performance Guarantees;

(ii) Redundancy – incorporate adequate redundancy of all unit processes and systems (i.e., capacity and/or number of units), to comply with the Specifications and ensure availability of all equipment sufficient to operate, monitor and meet the plant performance requirements at all times;

(iii) Maintainability – ensure safe access and isolation provisions such that all equipment, valves and instruments can be fully and safely maintained and calibrated without the need to shut down the plant, process unit or subsystems;

(iv) Staff training – enable operations staff to become familiar with the operations and maintenance aspects of such a plant; and

(v) Cost effectiveness – minimise the operation costs of the system as a whole, and each unit process individually.

P4.10.3 Sidestream N Removal Facility General Design Requirements

(a) The sidestream N removal facility shall be located within the C4B Working Area as indicated in the Drawings.

(b) The sidestream N removal facility shall be an anammox based system designed for the removal of ammonia from the incoming centrate and condensate streams.

(c) A flow equalisation/solids removal tank shall be provided to attenuate the impact of variable flows and high solids loading and provide the required buffer and attenuation for feeding the deammonification reactors. Settleable solids shall be removed in the flow equalisation/solids removal tank. The Contractor shall design the flow equalisation/solids removal tank to meet the requirements of the Deammonification Reactors.

(d) The daily volume of all flows directed to the Sidestream N removal facility and the schedule of dewatering operations shall be considered in sizing the flow equalisation/solids removal tank. The dewatering centrifuges are designed to operate on twenty-four (24) hours a day and seven (7) days a week basis. Duty rotation will be practiced, the design shall allow for centrifuge solids diversion to centrate on start-up and flushing to centrate on shutdown.

(e) The deammonification process shall be designed to selectively waste both inert material suspended in the feed stream as well as Ammonia Oxidising Bacteria (AOB) and Nitrite Oxidising Bacteria (NOB) populations. The intent of selectively wasting AOB populations is to maintain a beneficial ratio of AOB to anammox organisms. The goal of selectively wasting NOB organisms is complete washout. At the time of Tender, details shall be provided of the method of selective wasting.

(f) Discharge and selective NOB wasting from the sidestream N removal facility shall be directed to the discharge wet well and returned to the ILM's Grit Effluent Channel together with the Sidestream N removal effluent.

(g) Anammox organisms shall be retained within the process by selective wasting and returned to the deammonification reactors.

(h) Seeding of anammox organisms grown in the side-stream process shall be provided. Means specific to the technology selected shall be incorporated to facilitate operations to transfer seeding to other facilities.

(i) The sidestream N removal facility shall be fully automated to facilitate unattended operation. Major components within the facility shall be interconnected to provide a complete functional and automated system.

(j) Process control narratives shall be developed by the Contractor during detailed design for the automation and control of the equipment and devices. These will be subject to review and approval by the S.O.

P4.10.4 Seeding, Testing and Commissioning

(a) Requirements for seeding during plant testing and commissioning are detailed in Particular Specification P10 Plant Performance Testing.

P4.10.5 Dewatered Centrate Wet Well and Pumps

(a) The sidestream N removal facility shall receive flow from the centrate wet wells and pumps located in the Biosolids Building by Other contractors as shown in the Drawings.

P4.10.6 Flow Equalisation/Solids Separation Tanks

(a) Centrate pumps shall discharge into flow equalisation/solids separation tanks as shown on the Drawings. The inlet arrangement shall be designed to minimise turbulence.

(b) The use of gravitational flow from the flow equalisation/solids separation tanks to the deammonification reactors and discharge wet wells is encouraged for energy optimisation. An overflow shall be provided on each flow equalisation tank. The overflow shall gravitate to the discharge wet wells or plant drain system.

(c) The flow equalisation/solids separation tanks shall be equipped with a system to allow for periodic cleaning of accumulated solids from the tank's bottom. Settleable solids shall be removed using an automatic removal system. The reference design incorporates a tipping bucket system to flush settleable solids from the flow equalisation tank to the discharge wet wells.

(d) The flow equalisation/solids separation tanks shall be constructed as concrete tanks.

(e) If a tipping bucket system is used:

(i) The tipping bucket system shall be used when the flow equalisation tank water level is drawn down to flush any solids accumulated on the floor of the tank towards a sump for removal. The sump gravitates to the discharge wet well.

(ii) The tipping bucket system shall be suitable for flushing accumulated solids and debris from the flow equalisation tank floor with one flush. The flushing media shall be plant water from the G2D ring main.

(iii) The automatic tipping bucket flushing system shall operate as follows:

1) Flush is initiated, and the bucket is filled from G2D plant water ring main;

2) Once the bucket fills, it tips and its content spills along the floor of the flow equalisation tank. The velocity of the water washes solids into a hopper at opposite end of the tank. The bucket then returns to an upright position;

3) The flush water supply valve shall close once the bucket has tipped; and

4) Once the tipping bucket has tipped and on completion of a flush, the tipping bucket shall return to its normal resting position.

(iv) The tipping bucket shall be manufactured using type 316L stainless steel. The plate thickness shall be designed to withstand both the static and dynamic loading experienced under normal operation and installation conditions.

(v) The supports shall be fabricated entirely of type 316L stainless steel and anchored to the backwall of the equalisation tank.

(vi) The bearings shall be factory installed and tested and shall be permanently lubricated. The bearings may be occasionally submerged and shall be designed for this duty.

(f) The reference design incorporates a spray bar for foam suppression. The equalisation/solids separation tank design shall incorporate an automatic foam suppression system.

(g) The Contractor shall ensure that the structure is dimensionally correct to accept all selected equipment.

(h) A minimum of two(2) flow equalisation/solids separation tanks shall be designed to operate on an N-1 basis, where full hydraulic capacity and treatment performance at AA loads can be met with 1 tank out of service.

(i) The Contractor may propose a different type of flow equalisation/solid separation system. The Contractor shall provide details of the flow equalisation/solid separation system proposed in accordance with Volume 1 Annex N Schedule of Technical Data.

P4.10.7 Deammonification Reactors

(a) The Tender design is based on four (4) identical sequencing batch deammonification reactors (SBR). Irrespective of type, no fewer reactors shall be provided.

(b) The deammonification reactors shall be constructed as concrete tanks.

(c) The deammonification reactors shall be designed on N-1 basis, where full hydraulic capacity and treatment performance at AA loads can be met with 1 reactor out of service.

(d) Deammonification reactors shall include inter alia mixing, aeration system, decant system (where relevant) and selective sludge wasting system.

(e) Where mechanical mixers are provided, they shall be top entry high efficiency type with motors located above the reactors. Lifting davits shall be provided for maintenance. Identical mixers shall be used in all reactors to facilitate maintenance The mixers shall be designed for optimization of turn down to save on energy during the overall operation cycle.

(f) Mixer design shall comply with the General Specification G4.56 Tank Mixing Systems

(g) The Contractor shall complete and submit Computational Fluid Dynamics (CFD) analysis for proposed mixing and tank geometries to demonstrate, to the approval of the S.O., that the mixing effectiveness requirement is going to be achieved.

(h) Deammonification reactors shall be constructed to ensure free passage is maintained for transfer of surface scum and foam.

(i) Penstocks/valves shall be provided to enable each of the deammonification reactors to be individually isolated.

(j) A means of drainage of the deammonification reactors to the plant drain shall be provided.

(k) Each reactor shall be provided with at least three access openings with covers on the roof to allow access for personnel and equipment, such as temporary ventilation equipment, as required during maintenance activities. The opening dimensions shall be a minimum of 900mm x 900mm unless specified otherwise.

(l) Each reactor shall be provided with one access opening on the vertical wet well wall at the bottom of the reactor to allow access for personnel and equipment. The opening shall be a cast in 900 mm diameter pipe with a bolted blank flange. The blank flange shall be hinged and fitted with a lifting device to safely support the blank flange during removal and when the access way is open.

P4.10.8 Aeration System

(a) The oxygen requirements of the deammonification process shall be met by process aeration blowers. Blowers shall supply low-pressure air to the deammonification reactor diffused air systems.

(b) The blowers shall be located either indoor or installed under a roof structure.

(c) Each deammonification reactor shall be fitted with a floor mounted aeration grid complete with fine bubble diffusers, dropper pipes, manual isolation valve and modulating air flow control valve.

(d) The Contractor shall complete aeration calculations to demonstrate, to the satisfaction of the S.O., proper design of aeration requirements and equipment sizing. Note that the Contractor shall provide blowers with output capacity (with all duty blowers running) capable of meeting the total air flowrate required to simultaneously supply the SOTR to each deammonification reactor.

(e) The diffused air system shall be designed in accordance with General Specification G4.38 Fine Bubble Air Diffuser System. Other types of aeration diffusers shall be subject to S.O. approval.

(f) Blowers shall be designed in accordance with General Specification G4.10 Air Blowers. The blowers shall be of high-speed turbo air bearing type or high efficiency single-stage centrifugal blowers.

(g) Blower redundancy shall be as listed in Table P4. 12. The Contractor shall also provide space and connections for one(1) future standby blower.

Table P4. 12 Basis of Design: Sidestream N Blower Redundancy

PARAMETER	UNIT	DESIGN BASIS
Normal Operation Duty Units	#	N
Standby Units (minimum)	#	For N<4, N duty + 1 standby, For N≥4. N duty + 0.25N standby Where, N= Number of Duty Blowers; 0.25N shall be rounded up

(h) Suitable removable wire mesh guard shall be provided at all pipework within reach of personnel to prevent accidentally touching hot surfaces.

P4.10.9 Sludge Wasting

(a) Means shall be incorporated to allow selectively wasting of the sludge. Waste sludge (AOB and NOB bacteria) shall be discharged from the system while Anammox bacteria shall be returned to the deammonification reactor.

(b) The Contractor shall provide details of the method of selective wasting.

(c) Waste sludge is directed to the discharge wet wells and returned to ILM as shown on the Drawings.

(d) Provision shall be made to for transfer of Anammox sludge to other plants for seeding. In the reference design, diversion of Anammox sludge to the sludge wet well is provided to allow transfer of seeding to other plant via offloading to tanker if required.

(e) At least one (1) duty WAS pumps shall be provided for each deammonification reactor. At least one (1) common standby WAS pump shall be provided.

(f) A complete set of pipework, valves and fittings shall be provided for a fully functional system.

(g) The pumps shall be located indoor or sheltered. Selective wasting equipment such as hydrocyclones or micro-screens shall be sheltered.

P4.10.10 Discharge Wet Wells

(a) Two (2) discharge wet wells and one (1) sludge wet well shall be provided as shown on the Drawings.

(b) Each wet well shall be provided with a gravity overflow to the Sidestream N removal plant drain system which shall be connected to the Tuas Port Local Sewer (TPLS).

(c) The discharge wet wells shall be constructed as concrete tanks.

(d) The basis of design for the discharge wet wells is listed in Table P4. 13.

Table P4. 13 Basis of Design: Sidestream N Discharge Wet Wells

Parameter	**Unit**	**Value**
Number of Discharge Wet Wells (minimum)	#	2
Discharge wet well minimum hydraulic residence time (HRT), @ Maximum Month (MM) flowrate	hours	4 hours (Applicable for SBR only)
Number of Sludge Wet Wells (minimum)	#	1

(e) At least one (1) duty discharge transfer pump shall be provided for each wet well. At least one (1) common standby pump shall be provided. The Contractor shall also provide space and connections for one(1) future standby pump.

(f) A complete set of pipework, valves and fittings shall be provided for a fully functional system.

(g) The pumps shall be located indoor.

(h) Each wet well shall be provided with at least three access openings with covers on the roof to allow access for personnel and equipment, such as temporary ventilation equipment, as required during maintenance activities. The opening dimensions shall be a minimum of 900mm x 900mm unless specified otherwise.

(i) Each wet well shall be provided with one access opening on the vertical wet well wall at the bottom of the well to allow access for personnel and equipment. The opening shall be a cast in 900 mm diameter pipe with a bolted blank flange. The blank flange shall be hinged and fitted with a lifting device to safely support the blank flange during removal and when the access way is open.

P4.10.11 Chemical Dosing System

(a) The Contractor shall provide a complete Chemical Dosing System to meet the specific performance parameters in accordance with Volume 1, Annex O –Schedule of Process Guarantees. Chemical Dosing System shall be designed in accordance with General Specification G4.47 Chemical Plant. The controls of any chemical offloading and storage shall follow the similar system details and standard details in the P&IDs, including the instrumentation detailed in the local control station, LCS42.

(b) The designated chemical dosing storage and dosing area and layout shall be indicated in the Civil outline drawings.

P4.10.12 Plant Drainage System

(a) Plant Drainage System shall be provided to collect and convey emergency overflows and maintenance drain from the Sidestream N Removal process tanks and all process drains lines to a common plant drain header which shall be connected by gravity to the TPLS as shown in the Drawings.

P4.10.13 Odour Covers

(a) All process tanks shall be partially covered as a minimum, with the aim to utilise the area above the tank for other equipment such as blowers. The Contractor shall design the process tanks to ensure the open area coverage is adequate for natural ventilation. The partial covers shall be suitable to be odour control covers and the open areas shall be able to be fitted with odour covers without major modifications.

P4.10.14 Provision for Future OCF

(a) Space for a future Odour Control Facility (OCF) has been considered within the C4B Working Area in the Drawings. The Contractor shall include preliminary sizing and space for a future OCF based on Activated Carbon Units (ACU) and the Sidestream N removal process tank sizes. The allocated space for future OCF shall be indicated in the Civil outline drawings.

(b) The design of Sidestream N tanks shall include the provision to allow future installation of odour containment covers.

(c) The future OCF system for Sidestream N facility shall consist of the following as a minimum:

(i) Odour containment covers and associated foul air ducting and headers;

(ii) Pre-filters and OCF Fans;

(iii) Activated Carbon Units (ACU);

(iv) Discharge stack

(v) Other associated supporting systems.

(d) The preliminary design for future OCF system shall meet the following requirement :

Table P4. 14 Basis of Design: Future OCF System

Parameter	Unit	Value
Minimum air exchange rate from the headspace of process tanks *NOTE: For process tanks with aeration, the design aeration rate shall be considered for the odour extraction flowrate*	# per hour	6
Maximum outlet H_2S concentration at stack	ppmv	≤0.05
Maximum outlet odour concentration	OU	≤500
OCF system availability	%	100 (365 days/year)

P4.10.15 Other Requirements

(a) The following guidelines are included in the Appendices, guidelines are provided for information only. Final selection shall be approved by the S.O. prior to procurement and installation:

(i) Appendix P4B – Pipe Material Selection Guide; and

(ii) Appendix P4C – Valve Selection Guide.

P4.11 PLANT WATER

P4.11.1 Overview

(a) Plant Water (G2D) will be supplied at the contract boundaries for plant water use at the termination points shown in the Drawings and the Interface Tie-in Registers in Particular Specification P2 Appendices A and B. The termination points are off takes from the main G2D network (by Other contractors) which is designed to supply plant water at 300 kPa (g) @ 25.3 m SHD to various consumers sitewide.

(b) Plant water is supplied from the DLM MBR Filtrate. The water quality parameters are provided in Appendix P4A – Plant Water (G2D) Characteristics.

P4.11.2 Plant Water (G2D) Ring Mains

(a) Plant water (G2D) ring mains shall be provided within the plant area to supply plant water to various services and hose points as shown in the Drawings. The ring mains shall include a complete set of pipework, valves and fittings shall be provided for a fully functional system.

(b) Proposed plant water systems shall be subject to S.O.'s approval.

(c) Valves shall be provided in accordance with Appendix P4C– Valve Selection Guide.

(d) Interface flow and pressures shown in Interface Tie-in Registers in Particular Specification P2 Appendices A and B are based on flow contribution from all the interfaces. The Contractor shall design the system and G2D ring mains to ensure continuity of plant operation when any one of the interfaces are down for repair or maintenance.

(e) Where the pressure is insufficient or requirements not met, the Contractor shall provide booster pumps, including hydropneumatic tanks, where required, to prevent pressure fluctuations and frequent pump starts.

(f) Where the pressure is excessive and not safe for use in washing hoses, the Contractor shall provide suitable pressure regulating devices.

(g) The Contractor shall incorporate optimisation of plant water usage in the design and ensure the consumption is within the flowrate specified in Appendix P2B Interfaces for G2D Plant water.

(h) As described in earlier sections, the plant water going to first stage THP HS cooling heat exchanger shall undergo heat recovery from HS and will be reused for hot water supply to GWRF. This cooling water flowrate shall be included as part of the Plant Water consumption as well.

(i) Pipe sizes at Tie-in point shall be maintained. For THP, the Contractor shall provide information to C4A contractor for carrying out G2D network hydraulic analysis. Suitable water hammer and surge protection measures shall be provided based on the results of the analysis to prevent any potential damage to the plant and equipment or risk to safety.

(j) The Contractor shall provide adequate washing hose points for washing of the facilities and to dilute any minor spillages prior to disposal to the plant drain.

P4.12 MISCELLANEOUS PIPEWORK

(a) In addition to the DC and TCI interfacing pipes shown in the P&IDs, the pipelines shown in Table P4. 15 are required to convey fluids between the tie-in-points identified in the Interface Tie-in Registers in Appendix P2B. The proposed route for each pipeline is shown in the Drawings.

(b) A pipework system, complete with valves, fittings and supports shall be provided for each pipeline.

Table P4. 15 Miscellaneous Pipework

Descriptions	Fluid Code
ILM1 Primary sludge to PSI Holding Tank	PSI
ILM2 Primary sludge to PSI Holding Tank	PSI
ILM1 to WAS Holding Tank	WASI
ILM2 to WAS Holding Tank	WASI
Plant Drain and Recycles to ILM Headworks	PD
ILM 1 IW Tertiary Plant output to IW Storage	IW
ILM 2 IW Tertiary Plant output to IW Storage	IW

P4.13 AUTOSAMPLERS

(a) Automatic refrigerated samplers shall be provided at the indicative locations as shown in the Drawings. The configuration of the automatic samplers shall be subjected to the approval of the S.O.

P4.14 LIFTING EQUIPMENT

(a) The Contractor shall provide adequate lifting equipment to maintain all equipment provided as part of this Contract. The Contractor shall provide lifting davit(s) or electric traction elevator(s) to allow the safe transfer of heaviest components by Contractor's design from ground level to higher floors during operations and maintenance works. The lifting equipment proposed by the Contractor will be reviewed by the S.O. as part of the ALMOP study.

(b) Lifting equipment of various types and capacities shall be provided in accordance with the General Specification G4.3 Lifting Equipment.

(c) The Contractor shall, with the aid of the lifting equipment manufacturer/supplier, determine and confirm the parameters (e.g. load capacity ratings, dimensions, elevations, positions, safety settings, coverage area, maximum lift of hoisting hook, long travel speeds, cross travel speeds, lifting and lowering speeds, braking systems, headroom space, load test parameters, operational tests parameters, etc.) governing the crane, rails, corbels, inclined double-girder angle of inclination, wire rope, end-stops, safety features, etc. as necessary to ensure fully integrated, operable and safe lifting equipment able to lift, move and place all items that may be designated for overhauling from or onto the installed locations and designated laydown areas.

(d) 'Lifting' shall also implicitly refer to 'lowering'. This means that lifting equipment required to serve items of plant for all purposes related to overhauling shall also lower new, repaired, reworked, etc. items to facilitate installation or reinstatement.

(e) In conjunction with the preceding clause, the Contractor shall establish the weight of the heaviest equipment, valve, component, pipe fitting, ducting element, component, etc. to be lifted by any particular lifting device in question and ensure that the SWL for the lifting device is appropriately defined.

(f) All cranes provided to be dual speed cranes to ensure smooth movement during operations.

(g) Wireless remote controllers shall be provided on all motorised cranes. A backup wired control station shall also be provided.

(h) All travelling cranes shall be provided with permanent access platforms to enable the routine inspection and maintenance of the cranes. The access platforms shall be provided with lighting and power supply for maintenance equipment.

P4.15 AUTOMATIC LUBRICATION SYSTEM (ALS)

(a) The lubrication requirements of all mechanical equipment shall be fulfilled by Automatic Lubrication Systems (ALS).

(b) Each ALS shall consist of a controller or timer to run the system, reservoir(s) or canister(s) to store the lubricant, pump, metering valves or injectors, tubing and fittings to deliver the lubricant to the application points.

(c) The Contractor with the aid of the ALS suppliers and mechanical equipment systems suppliers shall select from the available types and configurations of ALS to cater to the lubrication requirements of all equipment requiring ALS taking particular note of the required ALS capacity and delivery pressure to convey the lubricant to all lubrication points.

(d) ALS of the single-line parallel configuration which are typically used for lubricating an individual machine shall not be used for the works.

(e) Selection of the ALS type and configuration shall be made from the following options:

Table P4. 16 ALS Types & Configurations

No.	ALS Type and Configuration	Operating Features	
1.	Dual-Line Parallel ALS	a.	ALS pump pressurises lubricant metering device via a 4-way, 2-position reversing valve and the first supply line.
		b.	Control piston of metering device shifts and directs the pressurised lubricant to the main piston which displaces lubricant to the application points.
		c.	Lubricant on the other side of the control piston is vented back to the reservoir through the second supply line and the other side of the reversing valve.
		d.	Shifting operation of the reversing valve enables the pump to pressurise the second supply line thereby repeating the cycle in reverse.
2.	Single-Line Progressive ALS	a.	Consists of various numbers of piston metering mono-block devices.
		b.	ALS pump provides a measured single shot, pulsed or continuous volume during lubrication cycle.
		c.	The first primed piston in the mono-block shifts, displacing lubricant to the bearing and diverting flow to the next piston. The second piston shifts and diverts flow to the third and so on to the last.
		d.	Mono-blocks are tubed from one to another to convey the lubricant from mono-block to mono-block.
		e.	Timer or pressure control feedback switch stops the ALS pump. Lubrication starts again according to the pre-set timing and duration of the specific lubrication regime.
3.	Centro-Matic	a.	ALS pump develops the lubricant pressure through a single main tubing to the injectors.
		b.	Each injector services one lubrication point and may be accurately adjusted to deliver the precise amount of grease or oil for each bearing.

(f) The ALS for each equipment or groups of equipment shall be completed with all accessories, instrumentations, alarms for low lubricant reservoir volume, fault alarms and appurtenances.

(g) ALS lubricant reservoir where applicable shall be sized to cater to 1 years' lubrication needs for all the equipment when operating at maximum plant output.

(h) The selected ALS capability shall include variable output flow adjustment for the lubricant pump, metering devices and injectors.

(i) For equipment where the above type of ALS is assessed to be not practical, ALS of the cartridge/canister type and/or single-line parallel configuration or equivalent which are typically used for lubricating an individual lubrication point may be considered subject to the approval of the S.O..

(j) The timer settings of the selected ALS shall be adjustable.

P4.16 ACCESS PLATFORMS, STAIRWAYS AND LADDERS

(a) Access platforms, stairways and ladders shall be provided to facilitate necessary passage access to plant locations requiring to be accessed and to safely access all items of plant including equipment, tanks, piping elements, valves and instruments. The platforms, stairways and ladders shall be designed taking into consideration any special access requirement that may be associated with all specific items of plant in question with respect to repair, maintenance, cleaning, greasing/lubricating, draining, replacement, adjustment, dismantling, hoisting operations, assembly, etc. The configuration of the platform, stairway and ladder shall be designed to suit each plant item and location requiring them.

(b) The Contractor shall include step-overs as and when required to maintain access to plant of for fire egress.

(c) The Contractor shall appoint a Professional Engineer to carry out the design of all relevant support elements and shall obtain approval from relevant statutory or Government authorities, such as BCA, after liaising with S.O. and Accredited Checker. The cost of design of system is deemed to have been included in the Contract Price.

(d) Access stairways shall be provided with yellow anti-slip nosing.

P4.17 GENERAL ACCESS AND INSPECTION COVERS FOR CONCRETE TANKS

(a) Access and Inspection covers shall provide the safe means of accessibility and/or inspection to items of plant that need to be accessed and/or inspected but which shall usually be covered for reasons of safety, security or restricted privileged access (e.g. only for skilled, trained, licensed or authorised personnel). These shall also be for keeping out undesirable entry of life forms which could disrupt process operations. The configuration of the access and inspection covers shall be designed to suit each plant item requiring them. All costs for access and inspection covers shall be deemed included in the tendered sum.

(b) All access and inspection covers should include a fall prevention grating. The grating should be located beneath the main cover, providing fall protection while the main cover is open for visual inspections or access to instrumentation. The grating shall be hinged and lockable. The grating shall be manufactured from either GRP or marine grade aluminium coated in a safety-yellow powder coat.

(c) Each access and inspection cover should automatically lock in the open position and be capable of being opened easily by a single operator.

P4.18 PLANT AND EQUIPMENT SUPPORT SYSTEMS

(a) The Contractor shall provide comprehensive support systems for all plant and equipment to the satisfaction of the S.O.. As far as is practical, support systems shall integrate supports for mechanical, electrical and I&C equipment, pipework and cabling. Such support systems are not shown fully in the BIM tender Model. The location and nature of all such support systems shall be based on the reference arrangement of Plant and equipment or as modified during the Contractor's detailed design. The Contractor shall be responsible for the detailed design of all support systems and the price for these is deemed to be included in the lump sum items in the relevant bills of quantities. Steelwork, and reinforced concrete used for supports will not be subject to measurement for payment.

(b) The Contractor shall appoint a Professional Engineer to carry out the design of all relevant support elements and shall obtain approval from relevant statutory or Government authorities, such as BCA, after liaising with S.O and Accredited Checker. The cost of design of support system is deemed to have been included in the Contract Price.

(c) The Contractor shall, where required provide pipe bridges / racks to provide support for multiple lines where multiple lines are required to transit through spaces that are not within the envelope of buildings; and provision of multiple, discrete supports / hangers is impractical. The Contractor will be required to design and fabricate the pipe bridge / rack.

(d) This Specification covers general and detailed design and layout requirements for process and utility pipe bridge / rack system and supplements the piping support / hanger specification.

(e) All lines shall be run side by side on pipe bridge / rack system to common supporting elevations for bottom of pipe

(f) Hot lines on pipe bridge / rack, shall be grouped and expansion loops shall be nested together. Hot lines should be grouped upon the top tier of the bridge /rack. The number of expansion loops shall be kept to a minimum.

(g) Lines handling corrosive fluids shall be run underneath lines handling non-corrosive fluids, typically the bottom tier of the bridge / rack and shall not where possible, be run overhead walkways or normal passages for personnel.

(h) Pipe bridges / racks with width greater than 15m shall be provided with minimum 10% spare space per layer, to be located at the midpoint of the layer as one continuous space. Future loading shall be considered on the 10% spare space. Below 15m width, the pipe bridge / rack shall include 20% spare space per layer / tier, with spare space located at midpoint of the layer as one continuous space.

(i) Consideration shall be given for instrument and electrical cable trays within the bridge/ rack where possible tray is to be kept on the uppermost tier or brackets designed as part of the portal frame, to accept tray work.

(j) Flange joints and placement of valves and instruments on lines, within the pipe bridge / rack shall be avoided wherever possible.

(k) If accessibility via mobile platform is not feasible, permanent access platform with ladder shall only be considered to allow access to valves, equipment, instruments etc.

(l) Where lines running on pipe bridge /rack system terminate at a point outside the battery limit of this contract package, these lines shall be provided with 250mm “green end “ / cutting allowance for field termination or tie in to the continuation of the line. Where such lines require the provision of a closure or valve at their battery limit, then this closure or valve will be within the supply of the upstream line.

(m) Where lines require to be tied into existing lines or lines outside the battery limit of this contract package, then the downstream line will be required to make the physical connection between the two lines.

(n) Pipe bridge / rack shall be designed such that lines are routed in shortest possible run with minimum number of fittings consistent with provision for expansion and flexibility. Assembly, removal and support of lines and equipment shall be taken into consideration.

(o) Pipe bridge / rack shall be designed to conform to “T”, “H”, or “Z” configuration. The shape chosen shall consider the area available.

(p) Pipe bridge / rack shall be 6m, 8m or 10m for single bay and 12m, 16m or 20m for double bay, and shall be limited to 4 tiers maximum. The spacing between pipe bridge / rack portals shall be taken as 6m in general, but this may be increased to 8m where site conditions or equipment layout beneath pipe bridge / rack dictates.

(q) Where pipe bridge / rack portal spacing exceeds the maximum self-supported span for a pipe or tray system, then intermediate support members shall be provided, to ensure adequate support.

(r) Minimum clearance beneath pipe bridge / rack shall be 2.50m minimum in both longitudinal and transverse directions. Road clearance shall be 7m minimum. Tier spacing shall permit man access.

(s) Footings for the pipe bridge / rack shall be provided as part of the pipe bridge / rack system.

(t) Minimum headroom between racks/tiers with metallic and non-metallic pipes shall be around 2.2m.

(u) Minimum headroom between racks/tiers with only chemical pipes shall be 0.5m.

(v) Pipe bridge / rack loads as a minimum consideration during design shall include:

(i) Dead load – piping, valves and load insulation;

(ii) Thermal load – load due to thermal expansion of piping and reaction force by internal pressure of expansion bellows;

(iii) Dynamic load - Load due to wind and seismic activity; and

(iv) Sustained load (live load) – liquid load for hydrostatic pressure test.

(w) Permanent access platform where required, shall be designed considering maximum design load for O&M personnel.

(x) The pipe bridge / racks shall be designed with adequate capacity to allow for the installation of temporary scaffold access platforms. The design load for the temporary access platform shall cater maximum O&M personnel load in addition to the platform self-weight.

(y) Pipe rack/bridge design shall include loadings of future pipes, cables, access platform (if required), O&M personnel (max of 3) and other loadings which needs to be considered to comply with code requirements.

(z) The Contractor shall include a superimposed dead load (SDL) of 2kPa on the top of each pipe rack to allow for future installation of Solar Panels.

(aa) Where pipe bridge /rack system is required to be run outside the battery limit of this contract package, the footings shall be provided by Other Contractor(s). The pipe bridge/ rack loads shall be provided to the Other Contractor(s) during the design.

(bb) Process lines are to predominantly be kept at lower tier, with utility and hot process at upper tier. Hot and cold lines should be kept at different tiers or failing this, at different groups within a tier.

(cc) Minimum pipe spacing between adjacent lines shall be decided based upon the outside diameter of bigger size flange with a minimum 300# rating to be considered. Outside diameter of the smaller pipe individual insulation thickness, and additional 25mm clearance. If flange is not present, minimum spacing shall be derived using this basis regardless. Actual line spacing especially at 90-degree bends and expansion loop locations, shall consider thermal expansion, contraction, non-expansion of adjacent line.

(dd) Large line sizes (above 350mm) shall be arranged close to portal columns to decrease beam bending moments.

(ee) Anchors on the pipe bridge/rack are to be distributed over two or three consecutive bays.

(ff) Anchors are to be provided within the structure immediately prior to the point where all hot lines leave the structure.

(gg) Expansion loops where required on high temperature lines shall be designed in accordance with ASME B31.3 and the loads from this, fed into the structural design basis. Expansion loops shall be designed around one column only. See Figure P4. 3**.**

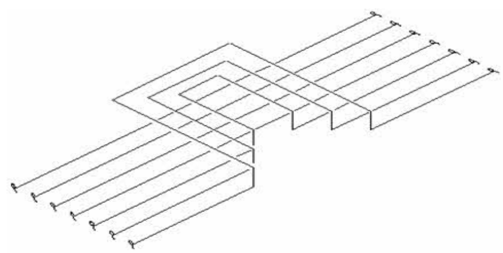

Figure P4. 3 Expansion Loops

(hh) Where stoplogs are provided in Sidestream N facility, the stoplog and its storage rack shall be stored within the same facility.

P4.19 MAJOR PIPE RACKS

(a) The Contractor shall provide the following major pipe rack(s) as described in Particular Specification P3 Civil and Building Works and the Drawings:

(i) Pipe Rack(s) conveying miscellaneous pipes as described in P4.12 from interface points with C4A and C3B1. Pipe arrangement on the pipe rack shall match the interconnecting pipe racks.

(b) The upper tier of the abovementioned major pipe rack(s) shall be provided for future pipes. The Contractor shall allow space and additional load capacity on the major pipe racks(s) for the following future pipes. This includes:

(i) 1 x TCI (Thickening Centrate Industrial), DN600

(ii) 1 x PD (Plant Drain), DN300

(iii) 2 x PSI (Primary Sludge Industrial), DN250

(iv) 2 x WASI (Waste Activated Sludge Industrial), DN250

(v) 1 x DC (Dewatering Centrate Industrial), DN250

P4.20 APPENDICES

(a) APPENDIX P4A – Plant Water (G2D) Characteristics

(b) APPENDIX P4B – Pipe Material Selection Guide

(c) APPENDIX P4C – Valve Selection Guide

(d) APPENDIX P4D – THP Feed Parameters

(e) APPENDIX P4E – LPHW supply requirements

(f) APPENDIX P4F – ROP/G1 Water Quality

END OF SECTION